

ABOUT THIS REPORT

This technical report was automatically generated by the Calyon Compressor Selection Software. The following tables comprehensively document the input parameters and calculation results for all steps of the selection process.

1. Project Information

Project Name	samet	
Description	-	
Creation Date	14.05.2026	
Report Reference	CAL-C0E6E1A1	

2. Compressor Specifications

Model Name	VX10HP	
Refrigerant	R-717 (NH3)	
Number of Cylinders	10	pcs
Maximum Speed	1500	RPM
Oil Charge	-	kg
Dry Weight	-	kg

3. Thermodynamic Conditions — Input Parameters

Evaporation Temperature	20	°C
Condensing Temperature	70	°C
Compressor Speed	1500	RPM
Superheat (Useful)	0	K
Superheat (Non-Useful)	5	K
Subcooling	0	K
Suction Line Pressure Loss	0	bar
Discharge Line Pressure Loss	0	bar

Calculated Pressures		
Evaporation Pressure	8.57	bar(a)
Condensing Pressure	33.12	bar(a)

4. Thermodynamic Performance Results		
Capacity & Power		
Cooling Capacity	1396.51	kW
Heating Capacity	1775.57	kW
Shaft Power	379.06	kW
Internal Compressor Power	360.10	kW
Efficiency Coefficients		
EER	3.68	
COP (Cooling)	4.68	
COP (Heating)	-	
Flow & Velocity Values		
Mass Flow Rate	5292.76	kg/h
Suction Volumetric Flow	810.98	m³/h
Discharge Volumetric Flow	293.07	m³/h
Suction Velocity	0.00	m/s
Discharge Velocity	0.00	m/s
Temperature & Pressure Results		
Suction Temperature	25.00	°C
Discharge Temperature	145.46	°C
Pressure Ratio	3.87	
Envelope Status	WARNING	

5. Motor Selection

Selection Criteria (Input)

Voltage	400	V
Safety Factor	1.1	
Manufacturer Filter	ALL	
IE Class	IE4	

Calculation Results (Output)

Required Motor Power	416.96	kW
Estimated Line Current	668.40	A
Min. Motor Power / Torque	417.0 kW / 2654.5 Nm	
Approx. Electrical Input	393.62	kW

Selected Motor Technical Specifications

Manufacturer	ABB	
Motor Type / Model	M3BP 315SMC 4	
Nominal Power	100	kW
IE Efficiency Class	IE4	
Nominal Speed	1491	RPM
Nominal Current	194	A
Torque	704	Nm

6. Mechanical Options

Compressor		
Crankcase Heater	350W	
Crankcase Heater (Alt)	2x 350W	
Heater Voltage	220-240V / 50-60Hz	
Base Frame	Base Frame for Concrete...	
Motor Support	Shims	
Capacity Control Voltage	24VDC	
Oil Level Switch	24V CALYON Switch	
Oil Cooler	Air	
Equalizing Line	Included	
Suction Safety Valve	Suction Safety Valve (Single,...)	
Internal Safety Valve	63 bar(g)	
Type Plate	Standard	
Stop Valves		
Stop Valves	Suction & Discharge w/...	
ASTM Valve Adapter	Not Included	
Mounting	Not Included	
Check Valves		
Discharge Check Valve	Not Included	
Check Valve Mounting	Not Included	
Coupling		
Coupling Model	CY105M	
Coupling Guard	Included	
Coupling Tools	Not Included	

Paint and Packing		
Paint	Jet Black RAL9005	
Spray Paint	Not Included	
Packing	Wooden Beam Packing	

7. Control Panel Options		
Control Panel		
Panel Type	CALYON Omni I/O Box (+ Disp...)	
Voltage/Hz	90-260VAC / 50-60Hz	
Display Language	English US	
Hardware	ABB	
Temperature Monitoring		
Temp Sensors	PT1000-2A (Grasso Stand...)	
Sensor Cable Length	5000 mm	
Cylinder Head Disch Temp Sensors	All Cylinders	
Thermo-Master Read-Out	Not Included	
Cyl Head Disch Temp Sensor Type	PT100	
Oil Temp Sensor	Not Included	
Pressure Monitoring		
Pressure Gauge Panel	Not Included	
Safety Switch(es)	Discharge Pressure S...	
HP Switches Type	Mechanical	

8. Part-Load Evaluation — Variable RPM

20,0 / 70,0 °C (ET / CT)

Speed (RPM)	Part Load (%)	No. of Cyl. (LP – HP)	Refrigerating / Heating Capacity		Power Req. (kW)	EER / COP
			Refrig. Cap. (kW)	Heating Cap. (kW)		
1500	100	10	1396,5	1775,6	379,1	3,68 / 4,68
1450	100	10	1350,0	1716,1	366,1	3,69 / 4,69
1400	100	10	1303,4	1656,7	353,3	3,69 / 4,69
1350	100	10	1256,9	1597,2	340,4	3,69 / 4,69
1300	100	10	1210,3	1537,8	327,5	3,70 / 4,70
1250	100	10	1163,8	1478,5	314,7	3,70 / 4,70
1200	100	10	1117,2	1419,1	301,9	3,70 / 4,70
1150	100	10	1070,7	1359,7	289,1	3,70 / 4,70
1100	100	10	1024,1	1300,4	276,3	3,71 / 4,71
1050	100	10	977,6	1241,1	263,6	3,71 / 4,71
1000	100	10	931,0	1181,8	250,8	3,71 / 4,71
950	100	10	884,5	1122,6	238,1	3,71 / 4,71
900	100	10	837,9	1063,3	225,4	3,72 / 4,72
850	100	10	791,4	1004,1	212,7	3,72 / 4,72
800	100	10	744,8	944,9	200,1	3,72 / 4,72
750	100	10	698,3	885,7	187,4	3,73 / 4,73
700	100	10	651,7	826,5	174,8	3,73 / 4,73
650	100	10	605,2	767,3	162,2	3,73 / 4,73
600	100	10	558,6	708,2	149,6	3,73 / 4,73
550	100	10	512,1	649,1	137,0	3,74 / 4,74
500	100	10	465,5	590,0	124,5	3,74 / 4,74
500	90	9	419,0	532,3	113,4	3,69 / 4,69
500	80	8	372,4	474,4	102,0	3,65 / 4,65
500	70	7	325,9	416,2	90,4	3,60 / 4,60

Speed (RPM)	Part Load (%)	No. of Cyl. (LP – HP)	Refrigerating / Heating Capacity		Power Req. (kW)	EER / COP
			Refrig. Cap. (kW)	Heating Cap. (kW)		
500	60	6	279,3	357,8	78,5	3,56 / 4,56
500	50	5	232,8	299,0	66,2	3,52 / 4,52
500	40	4	186,2	239,9	53,7	3,47 / 4,47